

LODI Outflow Boundary Condition: Reflection-Coefficient Validation Tests

Todd Harman

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1 Motivation

The LODI (Local One-Dimensional Inviscid) outflow boundary condition in `CCA/Components/ICE/CustomBCs/LODI2.cc` relaxes the boundary pressure toward a reference value p_∞ using a user-specified coefficient σ , following the NSCBC (Navier–Stokes Characteristic Boundary Conditions) formulation of Sutherland and Kennedy [1], itself building on Rudy and Strikwerda [3] and Poinso and Lele [2]. A larger σ anchors the boundary pressure more strongly (suppressing long-time drift away from p_∞) at the cost of reflecting more of any genuine outgoing disturbance back into the domain – the central trade-off in the NSCBC literature.

A review of `LODI2.cc` raised the question of whether a factor of 0.5 appearing twice in the code (once in the default characteristic-wave definitions L_1, L_5 , once again in the outflow relaxation term) constituted a double-counted, spurious halving of σ ’s effective strength. Reading the cited reference directly (Sutherland and Kennedy, Table 5 and Eq. 8–9) showed both factors of 0.5 are part of that paper’s own convention and are not duplicated – the code matches its cited source exactly. This document describes the follow-up validation effort: a suite of test problems that directly measures the boundary’s *reflection coefficient* $|R|$ as a function of σ , giving an empirical, reproducible check of the boundary’s behavior that is independent of the symbolic algebra review.

2 Background: the outflow relaxation term

For a subsonic outflow with the boundary-normal direction aligned with a coordinate axis, `LODI2.cc` sets the incoming characteristic wave amplitude (L_1 at a right/+ boundary, or L_5 at a left/– boundary) to

$$L_{\text{relax}} = \frac{1}{2L} \sigma c (1 - M^2) (p - p_\infty), \quad (1)$$

where c is the local speed of sound, M is the local Mach number, L is the domain length normal to the boundary, and p_∞ is the target (ambient) pressure – this is Sutherland and Kennedy’s Eq. 8–9 verbatim. $\sigma = 0$ reduces to a purely characteristic (non-reflecting) extrapolation; large σ pulls the boundary pressure rigidly toward p_∞ , which behaves increasingly like a reflecting wall. The tests below measure where a given σ actually sits on that spectrum.

3 Test methodology

3.1 Riemann-invariant decomposition at a probe

Each test initializes a small-amplitude Gaussian temperature spike at rest, off-center in the domain, using `customInitialization` $\rightarrow$ `gaussianTemperature`. A disturbance introduced this way decomposes into outward-traveling acoustic wave packets plus a stationary entropy anomaly that never reaches a distant probe. At a probe location on the domain’s symmetry line (defined precisely for each test family below), the probe’s `press_CC` and the velocity component normal to the boundary under test, u , are combined into right- and left-going Riemann invariants

$$p_+(t) = \frac{1}{2}(p(t) - p_\infty) + \frac{1}{2}\rho_0 c_0 u(t) \quad (\text{incident, right-going}), \quad (2)$$

$$p_-(t) = \frac{1}{2}(p(t) - p_\infty) - \frac{1}{2}\rho_0 c_0 u(t) \quad (\text{reflected, left-going}), \quad (3)$$

where ρ_0, c_0 are the ambient density and sound speed. The incident wave appears as a peak in $p_+(t)$ near the time the pulse first reaches the probe; the boundary’s reflection appears as a later peak in $p_-(t)$. The reflection coefficient reported for each test is

$$|R| = \frac{\max_t |p_-(t)|}{\max_t |p_+(t)|}. \quad (4)$$

This decomposition is exact only where the transverse velocity component is identically zero; both the 1D and 2D test families below are constructed so that the probe sits exactly on such a line.

3.2 On-the-fly probe extraction

The probe history is recorded by the `DataAnalysis` $\rightarrow$ `lineExtract` on-the-fly analysis module (`CCA/Components/OnTheFlyAnalysis/lineExtract.cc`), sampling `press_CC` and `vel_CC` at the probe cell every timestep into one small, continuously-appended text file.

The post-processing itself is `parametricStudies/postProcessTools/compute_LODI_reflectionCoeff.py`, which reads the `lineExtract` output directly (no further simulation queries needed), forms Eqs. (3)–(4), and writes $|R|$ to the parametric-study framework’s result file.

3.3 1D test family

`StandAlone/inputs/ICE/LODI_reflection/Lodi_reflectionCoeff_d{x,y,z}.ups` each use a long, thin domain (1D in the named axis; the other two directions are a single cell with symmetric boundaries) of length $L = 2.0$ m. The pulse sits at $x_{\text{spike}} = 1.2$ m; the probe sits at $x_{\text{probe}} = 1.6$ m (and analogously for y, z). Because the pulse is centered off-axis, it radiates two equal-amplitude planar wave packets, one toward each boundary. The far boundary is also LODI, so with

$$t_1 = \frac{x_{\text{probe}} - x_{\text{spike}}}{c_0} = 1.48 \times 10^{-3} \text{ s} \quad (\text{incident arrival}), \quad (5)$$

$$t_2 = \frac{2L - x_{\text{spike}} - x_{\text{probe}}}{c_0} = 4.44 \times 10^{-3} \text{ s} \quad (\text{reflection being measured}), \quad (6)$$

$$t_3 = \frac{x_{\text{spike}} + x_{\text{probe}}}{c_0} = 10.4 \times 10^{-3} \text{ s} \quad (\text{far-boundary contamination}), \quad (7)$$

($c_0 = \sqrt{\gamma(\gamma - 1)c_v T_0} = 270.12$ m/s for this test gas), the ordering $t_1 < t_2 \ll t_3$ gives a clean, uncontaminated window around t_2 in which to find the reflected peak.

3.4 2D test family

`Lodi_reflectionCoeff_2D_{xy,xz,yz}.ups` replace the 1D family’s planar pulse with a genuinely two-dimensional, radially-expanding (cylindrical) one: both in-plane spreads of the Gaussian are finite, so the wavefront that reaches the boundary under test is curved, striking it at a continuum of incidence angles rather than head-on. This directly exercises the transverse terms that the 1D LODI approximation in Eq. (1) drops.

The pulse and the probe both sit on the domain’s centerline in the *second* active axis (e.g., for the x-y plane test, both are at $y = L/2$); by the mirror symmetry of a centered pulse, the velocity component along that second axis is exactly zero along the whole centerline, which is what keeps the simple decomposition of Eq. (3) exact even though the wave itself is curved. All four in-plane boundaries are LODI; the domain size and pulse/probe placement ($L = 1.0$ m, spike at 0.6 m, probe at 0.8 m, both on the 0.5 m centerline) are chosen, via the same image-source construction used for Eq. (7), so that the two other boundaries’ reflections arrive safely after the one being measured:

$$t_1 = 7.40 \times 10^{-4} \text{ s} < t_2 = 2.22 \times 10^{-3} \text{ s} < t_{\text{other}} = 3.78 \times 10^{-3} \text{ s} < t_3 = 5.18 \times 10^{-3} \text{ s}. \quad (8)$$

The x-z and y-z plane variants are the identical construction with the in-plane axes relabeled; the x-z plane re-tests the x+ boundary (a cross-check of the implementation under a different domain orientation and patch decomposition), while the y-z plane tests the y+ boundary (a genuinely different boundary face). By the test gas’s isotropy, all three should – and, as reported below, do – land on the same $|R|(\sigma)$ curve.

4 Results

4.1 1D (planar incidence)

Table 1 gives $|R|(\sigma)$ for the 1D family. The x, y, and z variants are bit-for-bit identical at every σ , as expected since they are the same physics problem with the active axis permuted.

Table 1: 1D reflection coefficient vs. σ (x, y, z variants agree exactly).

σ	$ R $
0.00	0.01386
0.05	0.01415
0.10	0.01444
0.27 (default)	0.01546
0.50	0.01700
1.00	0.02116
2.00	0.04060

4.2 2D (curved incidence)

Table 2 gives $|R|(\sigma)$ for the 2D family. The x-y and y-z plane sweeps are bit-for-bit identical at every σ tested; the x-z plane was spot-checked at $\sigma = 0.27$ and also agrees exactly ($|R| = 0.026309$). This is a nontrivial cross-check: the y-z plane result in particular tests a genuinely different boundary face (y+ instead of x+) in a different domain orientation, yet the test gas’s isotropy still drives all three to the same curve.

Table 2: 2D reflection coefficient vs. σ (x-y and y-z plane sweeps agree exactly; x-z spot-checked at $\sigma = 0.27$).

σ	$ R $
0.00	0.02436
0.05	0.02467
0.10	0.02505
0.27 (default)	0.02631
0.50	0.02809
1.00	0.03255
2.00	0.04477

4.3 Discussion

Both families show the reflection coefficient increasing monotonically with σ , consistent with the trade-off described in Section 2: the default $\sigma = 0.27$ keeps $|R|$ to a few percent, while $\sigma = 2$ pushes $|R|$ toward 4–5% in both families.

The 2D family reflects systematically more than the 1D family at every σ – roughly 70% more at the default $\sigma = 0.27$ ($|R| = 0.0263$ vs. 0.0155). This is a real, physically meaningful result, not an inconsistency: the 1D LODI approximation underlying Eq. (1) is derived assuming a locally one-dimensional, normal-incidence wave, and drops the transverse pressure/velocity gradients that a genuinely curved wavefront has. The curved-wave test exercises exactly the terms the approximation omits, so a larger reflection coefficient there is the expected signature of that approximation’s known limitation, not a defect in the implementation.

5 Conclusions

- The LODI outflow boundary’s σ relaxation term behaves as the NSCBC literature predicts: reflection increases monotonically with σ , and the default $\sigma = 0.27$ keeps reflection to a few percent for normal-incidence (1D) waves.
- A genuinely curved (2D) wavefront reflects noticeably more than a planar one at every σ tested – an expected consequence of the 1D LODI approximation, confirmed quantitatively rather than assumed.
- Three independent 2D plane orientations, including two genuinely different boundary faces (x+ and y+), converge to the same $|R|(\sigma)$ curve, which is strong evidence against any axis-specific implementation bug.

References

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